📊 Customer Churn Prediction using Machine Learning

---

1. Introduction

Customer churn is a critical problem in the telecom industry, where customers discontinue services, leading to revenue loss. Predicting churn is challenging due to multiple influencing factors such as pricing, services, and customer behavior.

This project aims to develop a machine learning model to predict whether a customer will churn, enabling businesses to take proactive retention measures.

---

2. Objective

• Predict customer churn (Yes/No)
• Analyze customer behavior and patterns
• Handle imbalanced dataset
• Compare different machine learning models
• Improve churn prediction performance

---

3. Dataset Description

• Dataset: Telco Customer Churn Dataset
• Source: Kaggle / IBM
• Total Customers: 7043
• Features: 21

Key Characteristics:

• Mix of categorical and numerical data
• Moderate class imbalance
• Real-world telecom dataset

---

4. Data Preprocessing

4.1 Handling Missing Values

• Converted TotalCharges to numeric
• Handled missing values using appropriate methods

4.2 Feature Selection

• Removed customerID (irrelevant feature)
• Selected important variables for prediction

4.3 Encoding

• Converted categorical variables using Label Encoding

4.4 Train-Test Split

• Split dataset into training and testing sets

---

5. Handling Class Imbalance (SMOTE)

To address imbalance between churn and non-churn customers:

• Applied SMOTE (Synthetic Minority Over-sampling Technique)
• Balanced dataset for better model learning

Benefits:
• Improves recall
• Reduces bias toward majority class

---

6. Exploratory Data Analysis (EDA)

• Analyzed distributions of tenure, MonthlyCharges, TotalCharges
• Used countplots to analyze categorical features
• Identified key churn drivers:

- Low tenure → high churn
- High monthly charges → higher churn
- Contract type → strong impact

• Used correlation heatmap to detect relationships

---

7. Model Implementation

7.1 Logistic Regression

• Used for baseline classification
• Simple and interpretable

7.2 Random Forest

• Ensemble learning method
• Handles non-linearity well
• Achieved best performance

7.3 XGBoost

• Boosting algorithm
• Captures complex patterns
• Competitive performance

---

8. Model Evaluation

8.1 Evaluation Metrics

• Accuracy
• Precision
• Recall
• F1-score
• Confusion Matrix
• ROC Curve & AUC

8.2 Results (Best Model - Random Forest)

• Accuracy: ~78%
• ROC-AUC: ~0.83
• Recall (Churn): ~66%

Observations:

• Good overall performance
• Strong non-churn prediction
• Moderate churn detection

---

9. Model Comparison

Metric	Logistic Regression	Random Forest	XGBoost
Accuracy	Moderate	High	High
Recall	Moderate	Good	Good
ROC-AUC	Good	Best	Good
Performance	Baseline	Best Model	Competitive

---

10. Key Insights

• Tenure is a major factor in churn
• Monthly charges influence customer decisions
• Contract type significantly impacts retention
• Imbalanced data affects model performance
• Recall is more important than accuracy in churn prediction

---

11. System Implementation

• Built ML pipeline using Python
• Performed preprocessing, training, and evaluation
• Used visualization libraries for insights

---

12. Conclusion

The project successfully predicts customer churn using machine learning models. Random Forest achieved the best performance. The model provides valuable business insights for improving customer retention.

However, improving recall can further enhance real-world impact.

---

13. Future Work

• Improve recall using threshold tuning
• Use advanced models like LightGBM
• Deploy model using Flask/Streamlit
• Integrate real-time prediction system

---

14. References

• Kaggle Telco Customer Churn Dataset
• Scikit-learn Documentation
• XGBoost Documentation

---

15. Author

Ayush Radadiya
BCA Student | AI/ML Enthusiast